

(Ideas to keep in mind when solving problems)

- 3 Quantum numbers

$$(n, l, m) \rightarrow |n, l, m\rangle$$

n : Energy

l : total Angular momenta

m : "z" - component of Ang. momenta

$$\hat{H}|n, l, m\rangle = E_n |n, l, m\rangle$$

$$\vec{L}^2 |n, l, m\rangle = \hbar^2 l(l+1) |n, l, m\rangle$$

$$\hat{L}_z |n, l, m\rangle = \hbar m |n, l, m\rangle$$

$$E_n = -\frac{e^2}{2a_0} \frac{1}{n^2}$$

- Relationship between Quantum numbers:

$$m = -l, -l+1, \dots, 0, \dots, l-1, l$$

$$n = N + l + 1$$

N : called radial Quantum number

- Going from $|n, l, m\rangle$ to $\psi_{n\ell m}$:

In Dirac notation a general state is given by

$$|\psi\rangle = \sum_{n\ell m} c_{n\ell m}^{|} |n, l, m\rangle$$

$c_{n\ell m}^{|}$ are constants.

We then take this in the position-basis

$$\langle r, \theta, \phi | \psi \rangle \equiv \psi(r, \theta, \phi)$$

$$\left(\begin{array}{l} \text{and so} \\ \rightarrow \end{array} \right) = \sum_{n\ell m} c_{n\ell m}^{|} \underbrace{\langle r, \theta, \phi | n, l, m \rangle}_{\psi_{n\ell m}(r, \theta, \phi)}$$

or

$$\langle r, \theta, \phi | n, l, m \rangle = \psi_{n\ell m}(r, \theta, \phi)$$

- From all the maths we found

$$\psi_{n\ell m} = R_{n\ell}(r) Y_{\ell m}(\theta, \phi)$$

Note: Some textbooks use the notation $\psi_{n\ell m}$ or $\phi_{n\ell m}$ instead of $\psi_{n\ell m}$.

n	ℓ	m	$u_{n,\ell,m}$
1	0	0	$u_{1,0,0}(r, \theta, \phi) = \frac{1}{\sqrt{\pi}} a_0^{-3/2} e^{-r/a_0}$
2	0	0	$u_{2,0,0}(r, \theta, \phi) = \frac{1}{\sqrt{\pi}} (2a_0)^{-3/2} \left[1 - \frac{r}{2a_0} \right] e^{-r/2a_0}$
2	1	0	$u_{2,1,0}(r, \theta, \phi) = \frac{1}{2\sqrt{\pi}} (2a_0)^{-3/2} \frac{r}{a_0} e^{-r/2a_0} \cos \theta$
2	1	± 1	$u_{2,1,\pm 1}(r, \theta, \phi) = \mp \frac{1}{2\sqrt{2}\pi} (2a_0)^{-3/2} \frac{r}{a_0} e^{-r/2a_0} \sin \theta e^{\pm i\phi}$
3	0	0	$u_{3,0,0}(r, \theta, \phi) = \frac{1}{\sqrt{\pi}} (3a_0)^{-3/2} \left[1 - \frac{2r}{3a_0} + \frac{2}{27} \left(\frac{r}{a_0} \right)^2 \right] e^{-r/3a_0}$
3	1	0	$u_{3,1,0}(r, \theta, \phi) = \frac{2\sqrt{2}}{3\sqrt{3}\pi} (3a_0)^{-3/2} \frac{r}{a_0} \left(1 - \frac{r}{6a_0} \right) e^{-r/3a_0} \cos \theta$
3	1	± 1	$u_{3,1,\pm 1}(r, \theta, \phi) = \mp \frac{2}{3\sqrt{3}\pi} (3a_0)^{-3/2} \frac{r}{a_0} \left(1 - \frac{r}{6a_0} \right) e^{-r/3a_0} \sin \theta e^{\pm i\phi}$
3	2	0	$u_{3,2,0}(r, \theta, \phi) = \frac{1}{27\sqrt{2}\pi} (3a_0)^{-3/2} \left(\frac{r}{a_0} \right)^2 e^{-r/3a_0} (3 \cos^2 \theta - 1)$
3	2	± 1	$u_{3,2,\pm 1}(r, \theta, \phi) = \mp \frac{\sqrt{3}}{27\sqrt{\pi}} (3a_0)^{-3/2} \left(\frac{r}{a_0} \right)^2 e^{-r/3a_0} \sin \theta \cos \theta e^{\pm i\phi}$
3	2	± 2	$u_{3,2,\pm 2}(r, \theta, \phi) = \frac{\sqrt{3}}{54\sqrt{\pi}} (3a_0)^{-3/2} \left(\frac{r}{a_0} \right)^2 e^{-r/3a_0} \sin^2 \theta e^{\pm i2\phi}$

Table 7.2: Table of the hydrogen eigenfunctions for the principle quantum number $n = 1, 2, 3$. a_0 is the Bohr radius based on the reduced mass μ .

- Will use this table for problems.

Probabilities:

Recall

ψ (required)

$$\langle n, \ell, m | n, \ell, m \rangle = 1$$

$|\psi_{n\ell m}|^2 r^2 \sin \theta dr d\theta d\phi$: (Probability of finding the electron in a volume d^3r
 $d^3r = r^2 \sin \theta dr d\theta d\phi$)

Radial functions:

n	ℓ	$\mathcal{R}_{n,\ell}$
1	0	$\mathcal{R}_{10}(r) = 2a_0^{-3/2} e^{-r/a_0}$
2	0	$\mathcal{R}_{20}(r) = 2 (2a_0)^{-3/2} \left[1 - \frac{r}{2a_0} \right] e^{-r/2a_0}$
2	1	$\mathcal{R}_{21}(r) = 3^{-1/2} (2a_0)^{-3/2} \frac{r}{a_0} e^{-r/2a_0}$
3	0	$\mathcal{R}_{30}(r) = 2 (3a_0)^{-3/2} \left[1 - \frac{2r}{3a_0} + \frac{2}{27} \left(\frac{r}{a_0} \right)^2 \right] e^{-r/3a_0}$
3	1	$\mathcal{R}_{31}(r) = \frac{4\sqrt{2}}{9} (3a_0)^{-3/2} \frac{r}{a_0} \left(1 - \frac{r}{6a_0} \right) e^{-r/3a_0}$
3	2	$\mathcal{R}_{32}(r) = \frac{2\sqrt{2}}{27\sqrt{5}} (3a_0)^{-3/2} \left(\frac{r}{a_0} \right)^2 e^{-r/3a_0}$

Table 7.1: Radial wavefunctions of the hydrogen atom for the first three values for n . a_0 is the Bohr radius based on the reduced mass μ